LAB EXPERIMENT 8

Experiment 8: Stochastic POS Tagging Using a Probabilistic Model

Aim

To implement a simple stochastic part-of-speech tagger that assigns tags using probabilities estimated from training examples.

Algorithm

1. Start the program.
2. Import Counter from the collections module.
3. Prepare a small training set containing words and their POS tags.
4. Count how often each word occurs with each tag.
5. For each input word, choose the tag with the highest observed count (maximum estimated probability).
6. If a word is not in the training set, assign UNKNOWN.
7. Display the word and its predicted tag, then stop the program.

Code

```
from collections import Counter

training_data = [
    [("I", "PRON"), ("eat", "VERB"), ("apples", "NOUN")],
    [("You", "PRON"), ("eat", "VERB"), ("rice", "NOUN")],
    [("She", "PRON"), ("likes", "VERB"), ("apples", "NOUN")]
]

counts = Counter()
for sentence in training_data:
    for word, tag in sentence:
        counts[(word.lower(), tag)] += 1

def tag_word(word):
    candidates = {
        tag: count for (known_word, tag), count in counts.items()
        if known_word == word.lower()
    }
    if not candidates:
        return "UNKNOWN"
    return max(candidates, key=candidates.get)

sentence = ["I", "eat", "apples"]
print("Word -> Predicted Tag")
for word in sentence:
    print(word, "->", tag_word(word))
```

Output / Screenshot

```
data = [
    [("I", "PRON"), ("eat", "VERB"), ("apples", "NOUN")],
    [("You", "PRON"), ("eat", "VERB"), ("rice", "NOUN")],
    [("She", "PRON"), ("likes", "VERB"), ("apples", "NOUN")]
]

tag_counts = Counter()

for sentence in data:
    for word, tag in sentence:
        tag_counts[word.lower(), tag] += 1

def tag_word(word):
    possible_tags = {
        tag: count
        for (w, tag), count in tag_counts.items()
        if w == word.lower()
    }

    if possible_tags:
        return max(possible_tags, key=possible_tags.get)
    return "UNKNOWN"

words = ["I", "eat", "apples"]

for word in words:
    print(word, "->", tag_word(word))
```

```
I -> PRON
eat -> VERB
apples -> NOUN
```

Result

Thus, a simple stochastic POS tagging model was implemented by estimating word-tag frequencies from training examples and selecting the most frequent tag.